Case Study 31: String-Based Examination Answer Evaluation

1. Problem Statement

Develop an automated examination evaluation engine that compares a student's submitted string answers against an official answer key, applying positive marks for correct responses, negative penalties for incorrect responses, and neutral handling for unattempted questions.

2. Objectives

- Compare student string options against master answer keys with case-insensitive matching.
- Apply customizable marking schemes (e.g., +4 for correct, -1 for incorrect, 0 for unanswered).
- Generate a breakdown report detailing total score, correct attempts, wrong attempts, and skipped questions.

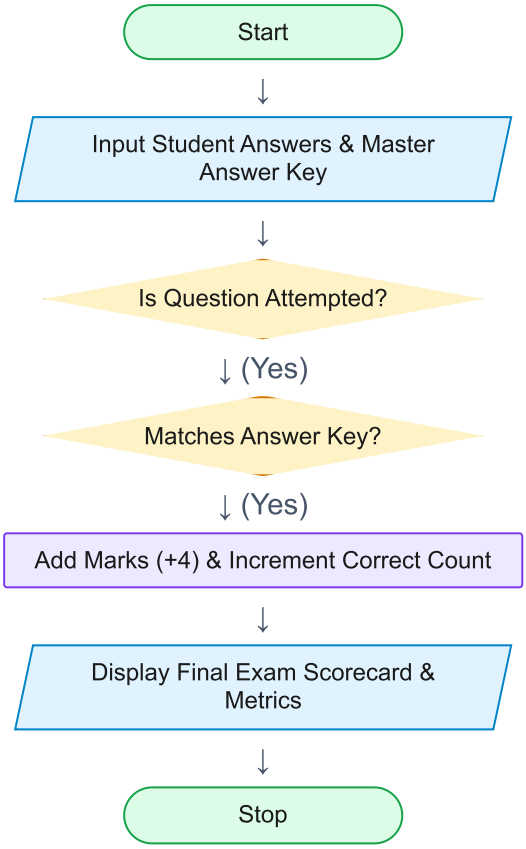
3. Assumptions

- Correct answer reward is set to **+4 marks** per question.
- Incorrect answer penalty is set to **-1 mark** per question.
- Unanswered questions (empty string or missing entry) receive **0 marks**.

4. Algorithm

1. Start.
2. Load answer key dictionary `(question_id: correct_option)` and student responses `(question_id: student_option)`.
3. Initialize `score = 0`, `correct_cnt = 0`, `wrong_cnt = 0`, and `unanswered_cnt = 0`.
4. For each question in the answer key:
    - Sanitize student response (strip whitespace and convert to uppercase).
    - If response is empty or unattempted:
        - Increment `unanswered_cnt` by 1.
    - Else if student response matches correct answer:
        - Add 4 to `score`.
        - Increment `correct_cnt` by 1.
    - Else (response mismatch):
        - Deduct 1 from `score`.
        - Increment `wrong_cnt` by 1.
5. Display score breakdown and evaluation summary.
6. Stop.

5. Flowchart



6. Pseudocode

```
START INPUT student_answers, answer_key marks_per_correct = 4 penalty_per_wrong = 1
score = 0 correct_cnt = 0 wrong_cnt = 0 unanswered_cnt = 0 FOR EACH q_id,
correct_ans IN answer_key DO student_ans = TRIM(UPPERCASE(student_answers[q_id]))
IF student_ans IS EMPTY THEN unanswered_cnt = unanswered_cnt + 1 ELSE IF
student_ans == UPPERCASE(correct_ans) THEN score = score + marks_per_correct
correct_cnt = correct_cnt + 1 ELSE score = score - penalty_per_wrong wrong_cnt =
wrong_cnt + 1 ENDIF ENDFOR PRINT "Total Score:", score PRINT "Correct Answers:",
correct_cnt PRINT "Wrong Answers:", wrong_cnt PRINT "Unanswered:", unanswered_cnt
STOP
```

7. Test Case and Expected Output

Test Case (5 Questions: Q1:A, Q2:C, Q3:B, Q4:D, Q5:A; Student attempted Q1:A, Q2:C, Q3:D, Q4:D, Q5:Skipped)

```
Examination Scorecard: - Total Score: 11 - Correct: 3 - Wrong: 1 - Unanswered: 1
```

8. Conclusion

The String-Based Examination Answer Evaluation system automates test evaluation, eliminating manual grading errors and delivering precise performance breakdowns according to custom scoring schemes.